3NF

PSS (CKi) ⟶ Nonprime

prime ⟶ Nonprime

if $\alpha \rightarrow \beta$

Both Nonprime ⟶ then not 3NF

⇒ has redundancy

Redundancy

STD → Dept Dept → Est

STD → EST Transitivity of Nonprime attributes.

R(A, B, C, D)

FD: A → B ⟶ Both side not Nonprim

$B \rightarrow C$ → Yes ||| Non prime

$C \rightarrow D$

$A \not\rightarrow B \not\rightarrow C \not\rightarrow D^+ = (A, B, C, D)$

So not in 3NF

$A^+ = (all)$

$PA = \{A\}$

$DA = \{B, C, D\}$

$\alpha = \phi$

No additional CK, A only CK.

$CK = A$

$PA = \{A\}$

$Non_PA = \{B, C, D\}$

$R(A, B, C, D, E, F)$

FDs: $AB \rightarrow CDEF$, $BD \rightarrow F$ (F: Non prime)

↳ not both side Non prime

$ABCDEF^+ = \{ \text{all} \}$

?

B prime
D Non prime
prime + Non prime = Non prime

↳ so not in 3NF.

$AB^+ = \{\text{all}\}$

CK = AB

PA = {A, B}

Non PA = {C, D, E, F}

DA = {C, D, E, F}

$\alpha = \phi$ → No other CK.

Attribute of CKs discovered so far

↳ Prime.

$CK_1 = AB$

$CK_2 = B\,C$

BCNF → 3.5 NF.

$\alpha \rightarrow \beta$ any Non trevial dependency.

R(A, B, C)

$A \rightarrow B \quad B \rightarrow C \quad C \rightarrow A$

ABC+ (all

$A^+ =$

$PA = \{A\}$

$DA = \{B, C\}$

$\alpha = \phi$

$B^+ =$

$C^+ =$

$CK_1 = A$

$CK_2 = B$

$CK_3 = C$

$\{A, B, C\}$

↳ Non existing cond^n

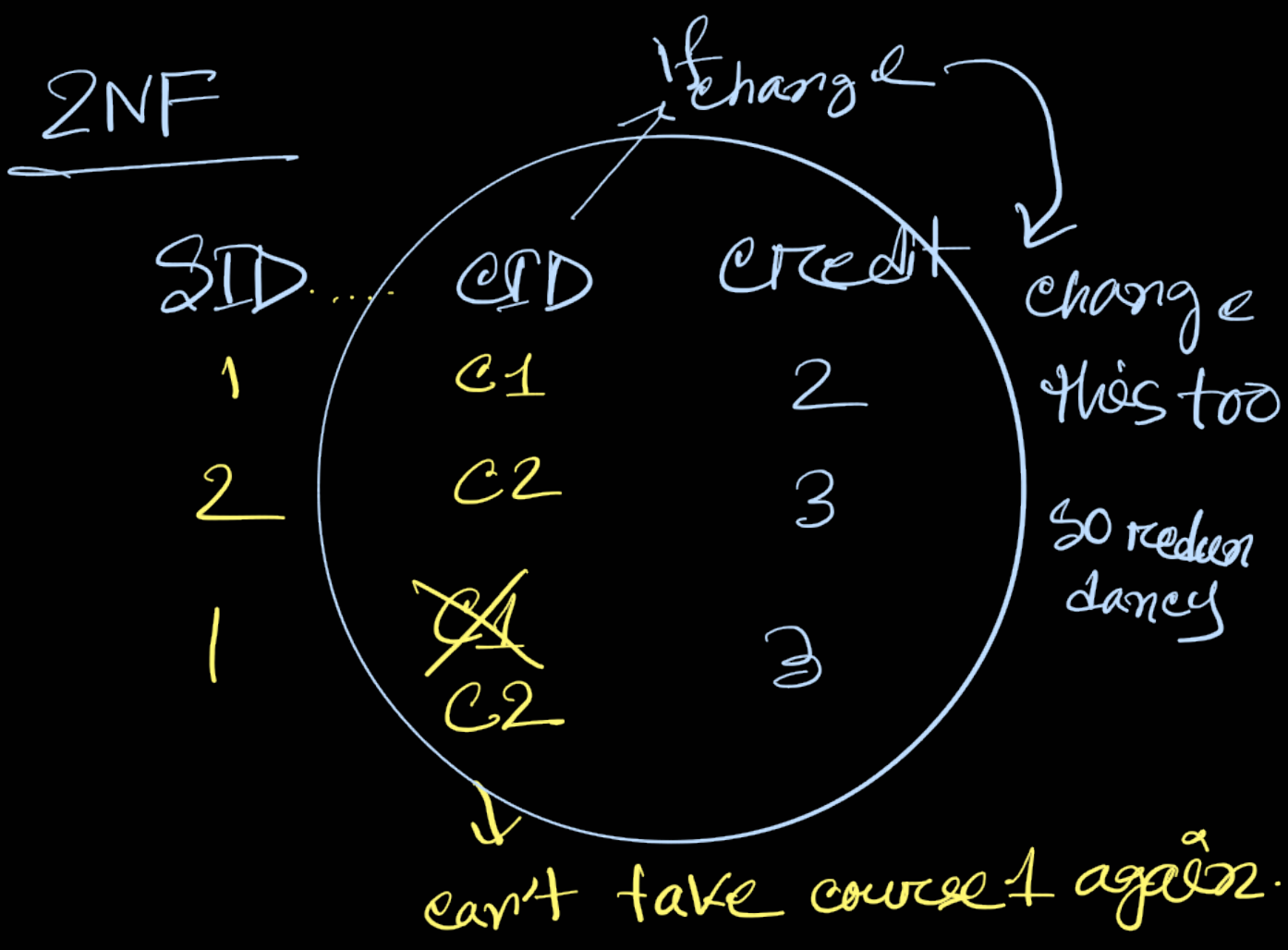
↳ Yes cond^n

↳ BCNF ✓

if $CK_3 = C$ wasn't present

→ Not BCNF.

2NF



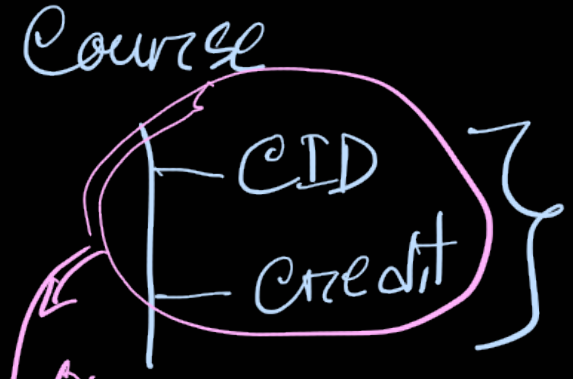
$CK = \{SID, CID\}$

part of pssv

CID → credit.

↳ nonprimev

PSS (CKi) → Non PA



so, make separate reference in database

Object boundary
↳ separate objects.

NonPA → NonPA

SID	Dept	Dloc
1	CSE	L2
2	EEE	L1
3	CSE	L2
4	CSE	L2

CK = {SID}

Dept → loc

Must be separate object / Entity
↳ Another condⁿ for NonPrime.